

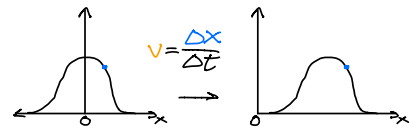
Transmission Lines

- circuits experience **Wave Phenomena** — at large distances, the **time delay** becomes significant
- Wave: transfer of energy through matter and/or space
 - Translation: energy (W) that is a function of time (t) and space (x) — where: $W(x, t=0)$ is some function of x : $f(x)$ — will have the relationship between time and space be: $\Delta x = v \cdot \Delta t$
 - this wave can be described by: $f(x - vt)$

- Temporal Wave Density: $\omega = \frac{2\pi}{T}$ — Number of cycles per [second] ← Period simplification
- Spatial Wave Density: $k = \frac{2\pi}{\lambda}$ — Number of cycles per [meter] ← Wavelength ($\beta = k$)

- Phase Velocity (v_p): velocity of point on wave

$$v = f \cdot \lambda = \frac{\omega}{k}$$
← time density per second
← space density per meter



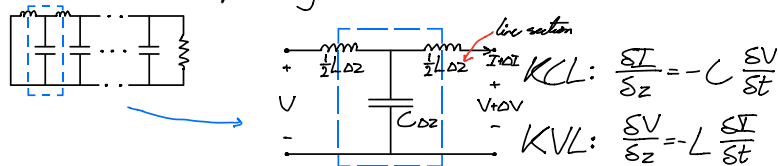
- Attenuation: conversion of the wave's energy into other energy forms as it passes through a medium
- Dispersion: a wave's phase velocity depends on its frequency
- Group Velocity (v_g): velocity of the variations in the shape of the wave's amplitude

$$v = \frac{\partial \omega}{\partial k} = \frac{c}{n(\omega) + \omega \cdot n'(\omega)} = \frac{c}{n_g(\omega)} \rightarrow \text{similar to } v = \frac{c}{n(\omega)}$$

- Wave Equation Differentiation: $W(x, t) = A \cdot f(x \pm vt)$

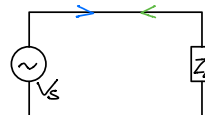
$$W'(x, t) \rightarrow \frac{\partial W}{\partial x} \mp \frac{1}{v} \cdot \frac{\partial W}{\partial t} = 0$$

- Lossless Transmission Lines (No dissipative elements)
 - Lumped-Element transmission line model for **lossless waves**
 - each capacitor must be charged to some value to allow the subsequent capacitor to charge — this creates a **time delay**
 - this model uses an infinitesimally small section of a transmission line



- Simple Waves on Transmission Lines
 - the load (Z_L) reflects back part of the wave (V_s) to create:

$$V(x, t) = V^+ f(x - vt) + V^- f(x + vt)$$



- the 2nd differentiation:

$$\frac{\partial^2 V}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 V}{\partial t^2} = 0 \xrightarrow{\text{solution}} V(z, t) = f_1(t - \frac{z}{v}) + f_2(t + \frac{z}{v}) = V^+ + V^-$$

- using Lumped-Element transmission line model equations:

$$\frac{\partial I}{\partial t} = -\frac{1}{L} \frac{\partial V}{\partial z} = \frac{1}{L v} [f_1' - f_2'] \xrightarrow{\text{solution}} I(z, t) = \frac{1}{L v} [f_1(t - \frac{z}{v}) - f_2(t + \frac{z}{v})] = I^+ + I^-$$

direction of wave propagation $\rightarrow$

wave velocity of: $v = \frac{1}{\sqrt{LC}}$

- Sinusoidal Voltages: $V(z, t) = |V_0| \cos(\omega t \pm \beta z)$

- Complex Instantaneous Voltage: $V(z, t) = \underbrace{V_0 e^{\pm j\beta z}}_{\text{Phasor Voltage}} \cdot e^{j\omega t}$

$$\beta = \frac{\omega}{v_p} = \frac{2\pi}{\lambda}$$

Phase Velocity $\rightarrow$

- Propagation Constant (γ): describes behaviour of EM wave along transmission line

$$\gamma = \alpha + j\beta$$

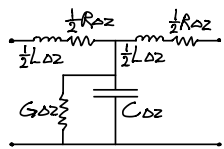
- Attenuation Constant (α): describes *decay* of signal amplitude [Np/m]

- Phase Constant (β): describes *periodic behaviour* of signal amplitude [rad/m]

- Lossy Transmission Lines

- a resistance (R) in the conductors resists the current

- a conductance (G) represents the leaked current through the dielectric between the conductors



$$KCL: \frac{\partial V}{\partial z} = -IR - L \frac{\partial I}{\partial t}$$

$$KVL: \frac{\partial I}{\partial z} = -GV - C \frac{\partial V}{\partial t}$$

$$\left. \begin{array}{l} KCL: \frac{\partial V}{\partial z} = -IR - L \frac{\partial I}{\partial t} \\ KVL: \frac{\partial I}{\partial z} = -GV - C \frac{\partial V}{\partial t} \end{array} \right\} \frac{\partial^2 V}{\partial z^2} = LC \frac{\partial^2 V}{\partial t^2} + (LG + RC) \frac{\partial V}{\partial t} + RG V$$

apply phasor notation

$$\rightarrow (R + j\omega L)(G + j\omega C) \cdot V$$

$\underbrace{R + j\omega L}_Z \underbrace{(G + j\omega C)}_Y$

$$\gamma = \sqrt{ZY}$$

$$V = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$\hookrightarrow \text{solution: } I = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

- Characteristic Impedance (Z_0)

- voltage to current ratio of a single wave propagating along an infinite line (no reflection)
- lossless transmission: $Z_0 = L v$

$$\text{lossy transmission: } Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} = \frac{Z}{Y} = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{(R + j\omega L)}{(G + j\omega C)}} = |Z_0| e^{j\theta}$$

- Reflected Waves

- Open Circuit: reflected wave's coefficient is *positive* — ("Free End")
- Short Circuit: reflected wave's coefficient is *negative* — ("Clamped End")

- Neper Unit [Np]: $1 dB = 20 \log_{10}(e) Np$

Low Loss Propagation

Instantaneous Voltage: $V(z,t) = V_0^+ e^{-\alpha z} \cos(\omega t - \beta z) + V_0^- e^{\alpha z} \cos(\omega t + \beta z)$

the Low-Loss Approximation requires: $R \ll \omega L \cap G \ll \omega C$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta$$

↳ this can be approximated using the binomial series to:

$$\alpha \doteq \frac{1}{2} \left(R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right) \quad \beta \doteq \omega \sqrt{LC} \left[1 + \frac{1}{8} \left(\frac{G}{\omega C} - \frac{R}{\omega L} \right)^2 \right]$$

Heaviside's Condition: when $R/L = G/C$, the line is said to be distortionless;

$$\alpha = \sqrt{RG} \quad \beta = \omega \sqrt{LC}$$

Power Transmission

Time-Averaged Power: $P = \frac{1}{2} \text{Re} \{ V_s \cdot I_s^* \} = \frac{1}{2} \frac{|V_0|^2}{|Z_0|} \cdot e^{-2\alpha z} \cdot \cos \theta$ [Np/m]

power attenuates (decays) at: $e^{-2\alpha z}$

power loss in decibels [dB]: $P_{\text{loss}} = 10 \cdot \log_{10} \left[\frac{P(0)}{P(z)} \right] = 10 \cdot \log_{10} (e^{2\alpha z})$

Power Fraction $\left(\frac{P(z)}{P(0)} \right)$: represents the fraction of the input power that reaches the output

Reflection and Standing Waves

EM waves reflect at the boundary between mismatched impedances

the superposition of all the waves within a transmission line creates standing waves

incident wave: $V_i = V_{oi} e^{-\alpha z} \cdot e^{j\beta z}$

reflected wave: $V_r = V_{or} e^{\alpha z} \cdot e^{j\beta z}$

Load Voltage: $V_L = V_{oi} + V_{or}$

Load Current: $I_L = I_{oi} + I_{or}$ Load Impedance

Reflection Coefficient: $\Gamma = \frac{V_{or}}{V_{oi}} = \frac{Z_L - Z_0}{Z_L + Z_0} = |\Gamma| e^{j\theta_r} \rightarrow V_{or} = \Gamma \cdot V_{oi}$ reflected voltage

Transmission Coefficient: $\tau = \frac{V_L}{V_{oi}} = 1 + \Gamma = |\tau| e^{j\theta_t} \rightarrow V_L = \tau \cdot V_{oi}$ transmitted voltage

Reflected Time-Average Power: $P_r = \frac{1}{2} \frac{|\Gamma|^2 |V_0|^2}{|Z_0|} \cos \theta$

the fraction of power reflected is: $\text{Re} \{ \Gamma \Gamma^* \} = |\Gamma|^2 = \frac{P_r}{P_i}$ reflected power [time-averaged] transmitted power

the fraction of power that is consumed by the load: $1 - |\Gamma|^2 = \frac{P_t}{P_i}$ transmitted power

- Voltage Standing-Wave Ratio

- the degree to which the voltage is divided between a *travelling wave* and a *true standing wave*

$$V_{ST}(z) = V_0 e^{-j\beta z} + \Gamma V_0 e^{j\beta z} \rightarrow \text{voltage wave containing both the forward and backward propagating waves}$$

$$V_{ST}(z) = \underbrace{V_0(1-|\Gamma|)e^{-j\beta z}}_{\text{travelling wave}} + \underbrace{2V_0|\Gamma|e^{j\Phi/2} \cdot \cos(\beta z + \Phi/2)}_{\text{standing wave}}$$

- *minimum* voltage occurs at: $z_{\min} = -\frac{1}{2\beta}(\Phi + (2m+1)\pi) ; m \in \mathbb{Z}$

- *maximum* voltage occurs at: $z_{\max} = -\frac{1}{2\beta}(\Phi + 2m\pi) ; m \in \mathbb{Z}$

- VSWR: $s = \frac{V_{ST}(z_{\max})}{V_{ST}(z_{\min})} = \frac{1+|\Gamma|}{1-|\Gamma|}$

- Wave Impedance ($Z_w(z)$)

- voltage to current ratio of superposition of all waves (incident and reflected) propagating along a finite line at location 'z'

$$Z_w(z) = \frac{V_{ST}(z)}{I_{ST}(z)}$$

- Input Impedance (Z_{in})

- the wave impedance at the input of a finite line (at $z=-l$)

- Conservation of Energy: incident power must equate to reflected power plus consumed power:

$$\frac{V_0^2}{Z_0} = \frac{\Gamma^2 V_0^2}{Z_0} + \frac{\tau^2 V_0^2}{Z_L}$$

- The Smith Chart: graphical method for finding various transmission line properties
- Input Impedance: combine all impedances of a circuit into one (Z_{in})
- depends on **location** of transmission line
- Derivation:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \xrightarrow[\substack{\text{normalise } Z_L \\ \text{to } Z_0}]{Z_L = \frac{Z_L}{Z_0}} \Gamma = \frac{z_L - 1}{z_L + 1} \rightarrow z_L = \frac{1 + \Gamma}{1 - \Gamma}$$

split z_L into **real** and **imaginary** parts ($z_L = r + jx$):

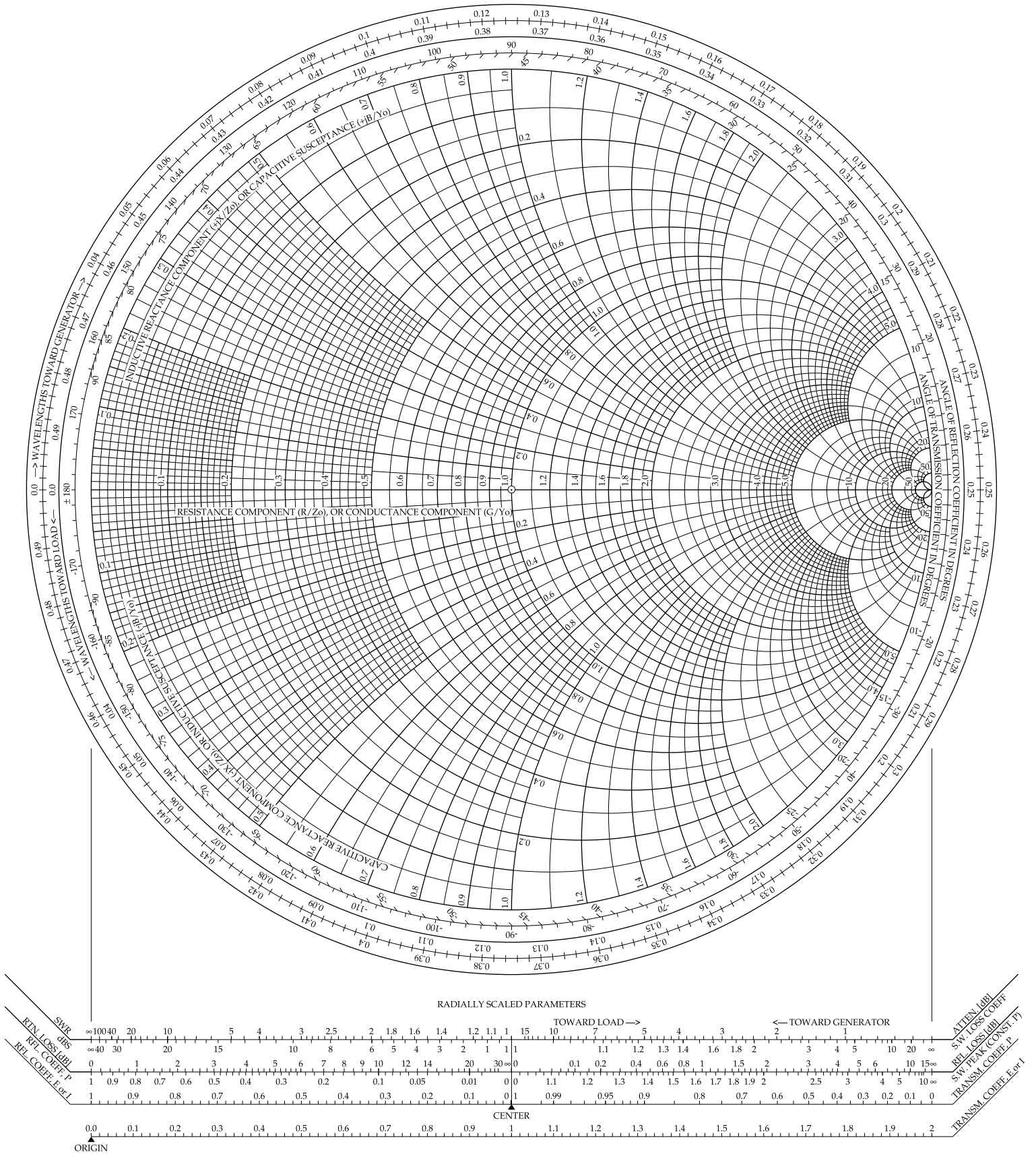
$$r = \frac{1 - \Gamma_r^2 + j\Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}, \quad x = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$\hookrightarrow \left(\Gamma_r - \frac{r}{1+r}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r}\right)^2, \quad \left(\Gamma_r - 1\right)^2 + \left(\Gamma_i - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$$

- this equation describes a family of circles, each associated with a specific " r "
- this equation describes a family of circles, each associated with a specific " x "
- the Wavelength rings represent fractions of a wavelength (λ)
 - every 360° around the Chart represents half a wavelength.
- Admittance: $y = \frac{1}{z}$
 - any normalised wave impedance (z_w) is 180° from its admittance (y_w)
- Chart Functionality
 - using a known impedance on the transmission line (z_w):
 1. move towards the **input impedance** by rotating **clockwise** ($\rightarrow z_{in}$)
 2. move towards the **load impedance** by rotating **counter-clockwise** ($\rightarrow z_L$)
 - this can be used for $z_L \rightarrow z_{in}$ or $z_{in} \rightarrow z_L$
- the reflection and transmission coefficients can be found by:
 1. drawing a line from the centre to z_L : Γ
 2. drawing a line from the origin to z_L : τ
 - for Magnitude: compare the physical length to the respective **Parameter** sections below the Chart
 - for Angle: continue the line from z_L to the edge of the Chart and check the respective **Angle of ring** for an intersection

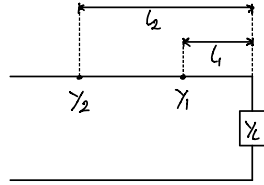
The Complete Smith Chart

Black Magic Design



- **Stub Matching:** connect an open circuited or short circuited transmission line to the main transmission line to alter the load impedance

1. Finding Stub Location:
 - a) draw **circle** centred at **centre** of Smith Chart of radius z_L or y_L

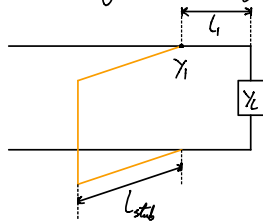


- b) rotate towards generator or load until intersection with **unit circle** ($r=1$ from centre)

- c) first intersection is γ_1 , second is γ_2

- d) determine equivalent wavelength of γ_1 & γ_2
 $\gamma_1 \rightarrow l_1$ $\gamma_2 \rightarrow l_2$

2. Determining Stub Length:



- a) find the **opposite** imaginary circle on Smith Chart to the circle that corresponds to the **stub location**
 $\gamma_{stub} = 0 + jA$

- b) rotate from $0 + j0$ — open circuit, or $\infty + j\infty$ — short circuit, (towards generator or load) to the wavelength corresponding to γ_{stub}
 $\rightarrow l_{stub}$

- the added **stub** will eliminate the line **susceptance**: $\gamma_1 + \gamma_{stub} = 1 + j0$

Electromagnetic Waves in Free Space

- **Transverse Electromagnetic Waves**
 - both **Electric** and **Magnetic** fields exist perpendicular to wave propagation

$[E] = V/m$ Time Averaged Power: $\langle S \rangle = \frac{1}{2} \text{Re} \{ \vec{E} \times \vec{H} \}$
 $[H] = A/m$ Poynting Vector: $[S] = W/m^2$

- **Physical Interpretation of Maxwell's Equations**

$\nabla \cdot \vec{D} = \rho_v$ • electric field lines diverge from a charge

$\nabla \cdot \vec{B} = 0$ • positive and negative magnetic charges always exist together preventing magnetic field lines from diverging

$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ • magnetic field lines curl around a current density and also curl if there is a change in electric flux density

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ • electric field lines curl if the magnetic flux density is changing

- Wave Equation of EM waves
 - using Faraday's and Ampere's Law, the coupled rotation of EM waves can be described with a partial differential equation:

$$\left. \begin{aligned} \nabla \times \vec{H} &= \epsilon \frac{\partial \vec{E}}{\partial t} \\ \nabla \times \vec{E} &= -\mu \frac{\partial \vec{H}}{\partial t} \end{aligned} \right\} \nabla^2 \vec{E} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0 ; \vec{E}(x, y, z, t), \vec{H}(x, y, z, t)$$

- Phasor Representation: separate the *spatial* and *temporal* components:

$$\vec{E}(x, y, z, t) = \vec{E}_s(x, y, z, \omega) e^{j\omega t}$$

- 1D Complete Wave Representation:

- a wave propagating along the *z* direction has electromagnetic components: E_{xs}, H_{zs}

$$E_{xs} = E_{x0}^+ e^{-jk_0 z} + E_{x0}^- e^{jk_0 z} \quad k_0 = \omega \sqrt{\epsilon_0}, \quad jk = \alpha + j\beta$$

$$H_{zs} = \frac{E_{x0}^+}{\eta_0} e^{-jk_0 z} + \frac{E_{x0}^-}{\eta_0} e^{jk_0 z} \quad \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

- Time-Averaged Power Density: $\langle \vec{S} \rangle = \frac{1}{2} \text{Re} \{ \vec{E} \times \vec{H}^* \}$

- Conductive Current Density ($\vec{J}_{\text{conductive}}$)

- Ampere's Law: $\nabla \times \vec{H} = \vec{J}_{\text{source}} + \vec{J}_{\text{conduction}} + \vec{J}_{\text{displacement}}$

- in conductive materials both source and conduction current density are present:

$$\nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t} + \sigma \vec{E} = j\omega \epsilon \vec{E} + \sigma \vec{E}$$

- represent current density with an *effective permeability*: $\epsilon_{\text{eff}} = \epsilon - j \frac{\sigma}{\omega} = \epsilon' - j\epsilon''$

- Intrinsic Impedance (η) becomes: $\eta = \sqrt{\frac{\mu}{\epsilon_{\text{eff}}}}$

- Skin Depth (δ): most current travels along outer edge in a *good conductor*: $\sigma \gg \omega \epsilon$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}}, \quad \alpha + j\beta = (1+j) \sqrt{\frac{\mu\omega\sigma}{2}}$$

$$\hookrightarrow \delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\mu\omega\sigma}}$$

- Reflections
 - the amount of reflection at a boundary between materials depends on conductivity of materials

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \rightarrow \mathcal{R} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

- Standing Wave occurs at $\Gamma = -1$; η_2 is a perfect conductor

$$E_{x1}(x, t) = 2E_{x0} \sin(\beta z) \sin(\omega t)$$

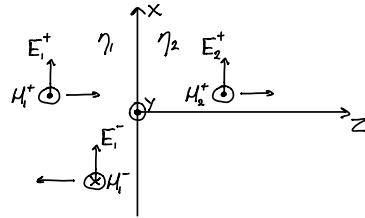
- Normal Incidence Reflections

- EM waves interacting with boundaries at normal incidence angle

$$\text{incident: } E_{xs1}^+ = E_{x01}^+ e^{jk_1 z}$$

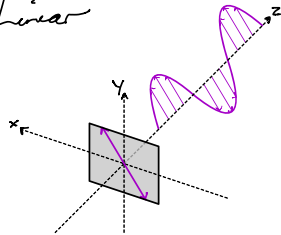
$$\text{reflected: } E_{xs1}^- = E_{x01}^- e^{jk_1 z}$$

$$\text{transmitted: } E_{xs2}^+ = E_{x02}^+ e^{jk_2 z}$$



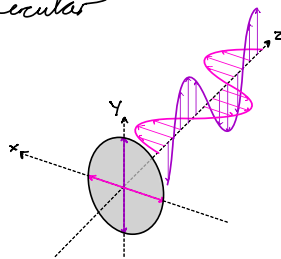
- Polarisation
 - geometrical orientation of the oscillations of a transverse wave

- Linear



$$\vec{E}(z, t) = \text{Re} \langle E_x e^{j(\omega t - \beta z)}, E_y e^{j(\omega t - \beta z)}, 0 \rangle$$

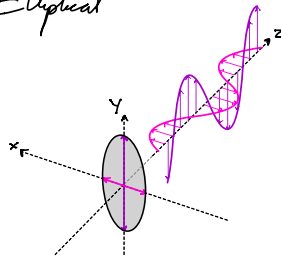
- Circular



- when $\Phi = 90^\circ \rightarrow$ Left Circular Polarisation
- when $\Phi = -90^\circ \rightarrow$ Right Circular Polarisation

$$\vec{E}(z, t) = \langle E_x \cos(\omega t - \beta z), E_y \cos(\omega t - \beta z + \Phi), 0 \rangle$$

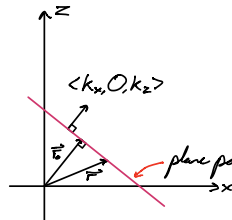
- Elliptical



$$E_x \neq E_y$$

$$\vec{E}(z, t) = \langle E_x \cos(\omega t - \beta z), E_y \cos(\omega t - \beta z + \Phi), 0 \rangle$$

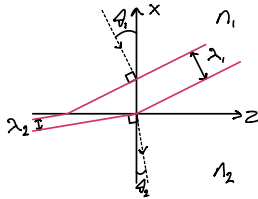
- Plane Waves
 - wave with constant value across plane at any point along propagation direction
 - Electromagnetic Wave Propagation can be thought of as travelling *plane waves*



$$(\vec{r} - \vec{r}_0) \cdot \vec{k} = 0 \rightarrow |\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

$$\vec{E} = \vec{E}_0 e^{j(\omega t - \vec{k} \cdot \vec{r})}$$

- Plane Wave travelling along $\hat{k}$: $\hat{k} \times \vec{E}_0 = \sqrt{\frac{\mu}{\epsilon}} \vec{H}_0$
- Wave Dynamics
 - behaviour of *plane wave* at boundary between two mediums (n_1/n_2)



$$\lambda_1 = \frac{c}{f \cdot n_1} \quad \lambda_2 = \frac{c}{f \cdot n_2}$$

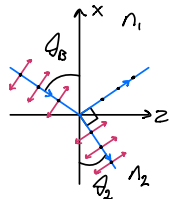
$$\frac{\lambda_1}{\sin \theta_1} = \frac{\lambda_2}{\sin \theta_2} \rightarrow n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- Critical Angle
 - incidence angle (θ_1) corresponding to refracted angle (θ_2) of 90°
$$\theta_1 = \theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right); n_1 > n_2$$

- Total Internal Reflection ($\theta_1 \geq \theta_c$)

- Evanescent Wave: exponentially decaying wave enters the second medium in the event of total internal reflection

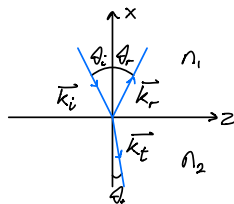
- Brewster's Angle (θ_B)



- reflected wave has zero electromagnetic energy; entire incidence wave is transmitted

$$\tan \theta_B = \frac{n_2}{n_1}$$

- Reflection & Transmission Coefficients

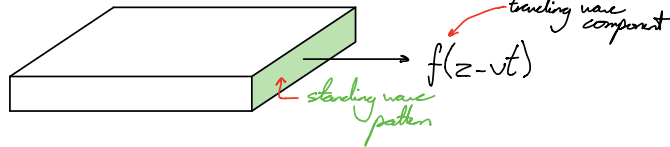


$$\Gamma = \frac{n_2 \cos \theta_t - n_1 \cos \theta_i}{n_2 \cos \theta_t + n_1 \cos \theta_i}$$

$$\mathcal{T} = 1 + \Gamma$$

Waveguides

- restricts a transverse **standing wave's** propagation along a certain direction
- waveguide allows wave to "bounce" around in the general direction of the guide, whereas transmission line restricts propagation along direction of the line



- EM waves can be separated into normal and transverse components:

$$\vec{E} = \vec{E}_\perp + \hat{z} E_z$$

$$\vec{H} = \vec{H}_\perp + \hat{z} H_z$$

$$\vec{\nabla} = \vec{\nabla}_\perp + \hat{z} \frac{\partial}{\partial z}$$

- Helmholtz Equations

- develop equations for separated electric and magnetic wave propagation:

Transverse Modes

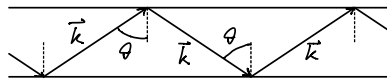
TE wave: $[\nabla_\perp^2 + (k - k_z)^2] H_z = 0$; no $\vec{E}$ field along $\hat{z}$, s-polarisation

TM wave: $[\nabla_\perp^2 + (k - k_z)^2] E_z = 0$; no $\vec{H}$ field along $\hat{z}$, p-polarisation

$$H_z = e^{-jk_z z} \cdot G \cos(k_x x) \cos(k_y y); \quad k_x^2 + k_y^2 + k_z^2 = k^2 = \omega^2 \mu \epsilon$$

- Waveguide Modes

- wave propagation only occurs at discrete angles of incidence (θ) within waveguide



- Cutoff Frequency (ω_c)

- each mode has a **minimum frequency** for propagation:

$$\omega_c = \frac{1}{\sqrt{\mu \epsilon}} \cdot \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}; \quad k_x = \frac{m\pi}{a}, \quad k_y = \frac{n\pi}{b}$$